

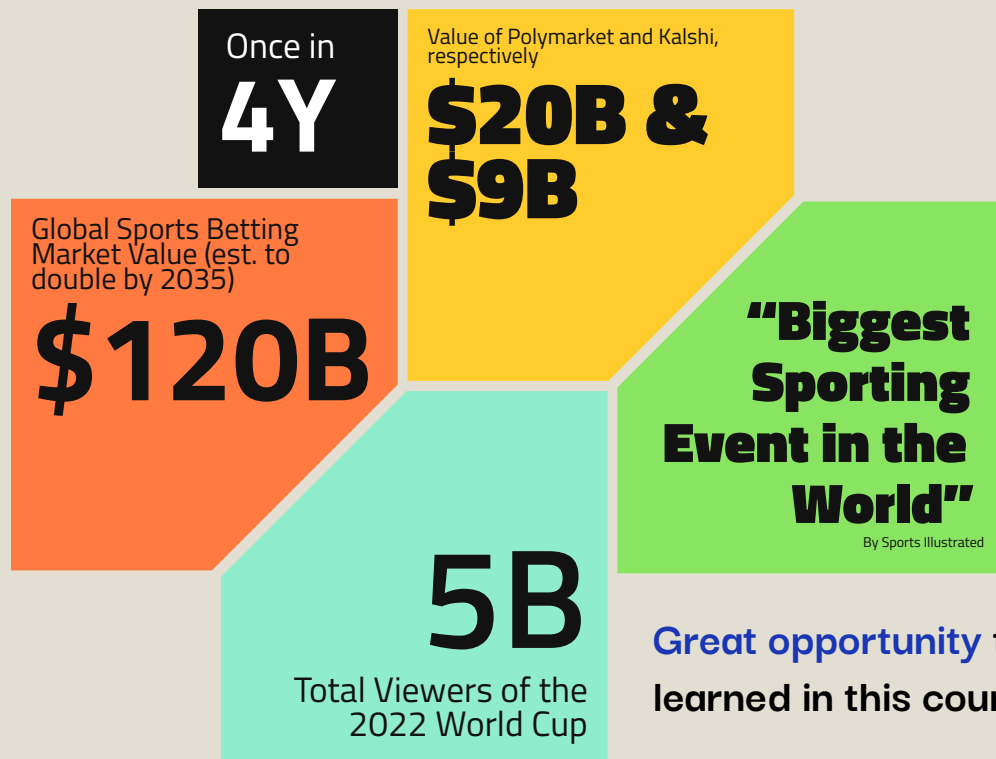
FIFA 2026 World Cup Predictor

DataSci 207, Final Presentation

Chase Foster
Leah Finger
Nicole Yokota
Kevin (Shinkyum) Rho
Jay Yao



Why predict the match results of the 2026 World Cup?



Great opportunity to apply what we've
learned in this course!

Related Work: Feature availability depends on the goal

- **Two common paradigms**
 - Post-match / in-match indicators → explain why teams win
 - Pre-match features → forecast upcoming matches
- **Song et al. (2024): post-match, coaching insight**
 - Goal: identify the technical dimensions that are most predictive of outcomes
 - Binary task: win vs. no win
 - Best model: ANN
 - Strength: SHAP identifies most influential features: *On Target, Shooting Opportunity, Ball Progressions*
 - However, indicators aren't available pre-kickoff
- **Ramadhani et al. (2026): pre-match forecasting**
 - Features: FIFA rank/points differences + squad strength differences + match context
 - 3-class outcome: Win/Draw/Loss
 - Best model: HistGBM
 - Challenge: very low draw recall (~0.08); squad metrics may limit reproducibility
- **Our takeaway**
 - We recognize that accurately predicting draws will likely be a challenge.
 - Building upon Ramadhani et al., we mindfully engineer reproducible, pre-match features and introduce dynamic form features.

Our Dataset

International soccer match results (1872 ~ 2026)

Primary Dataset (source: Kaggle)
49,071 total records

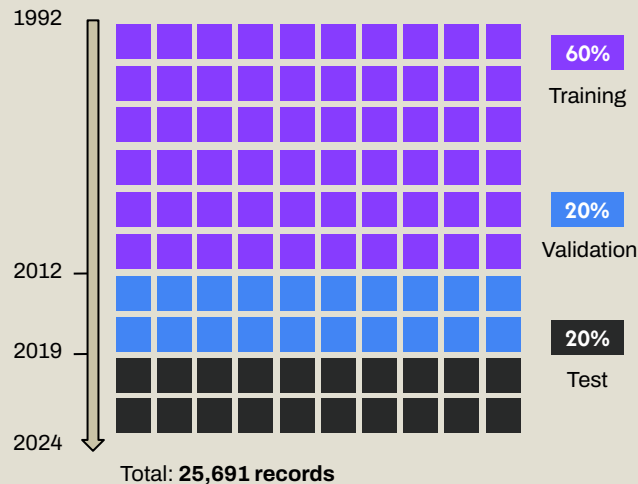
FIFA rankings (1992 ~ 2024)

Supplementary Dataset (source: Kaggle)
13,130 total records

Merge &
Process



Our Dataset



Our Features

We have engineered **15** features that can best describe the team's strength (traditional and recent) and the match details.

Captures the traditional strength of a team

- » ELO difference/sum
- » FIFA Rank difference/sum
- » H2H Team 1 wins / Team 2 wins / Draws

Captures the recent strength of a team

- » Points difference/sum for last 5 games
- » Goals difference/sum for last 5 games
- » Number of games team 1 or 2 played (max 5)

Captures the match details

- » Is it a friendly or a competitive match?
- » Is it played in a neutral ground?

Our Outcomes

Class 1: Draws



Class 2: Team 1 wins

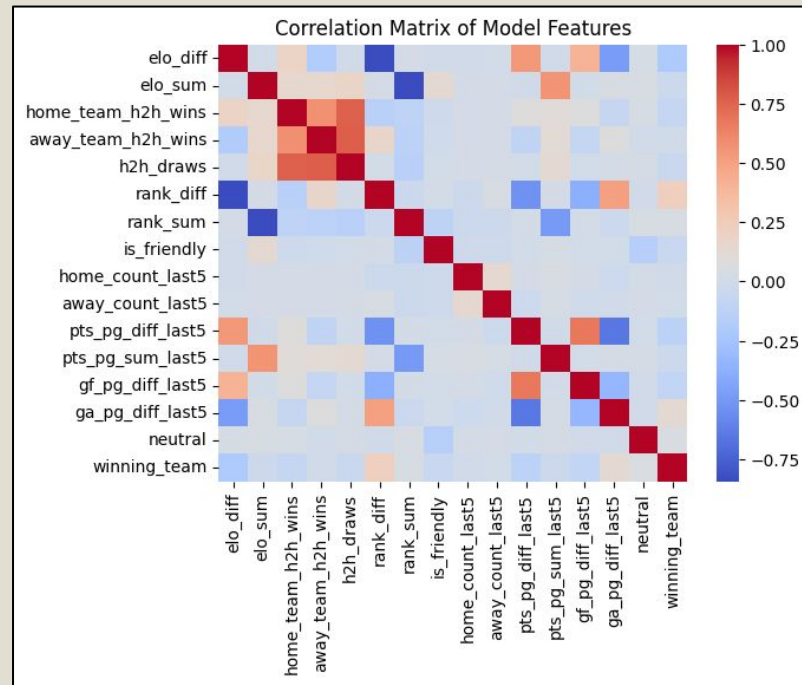
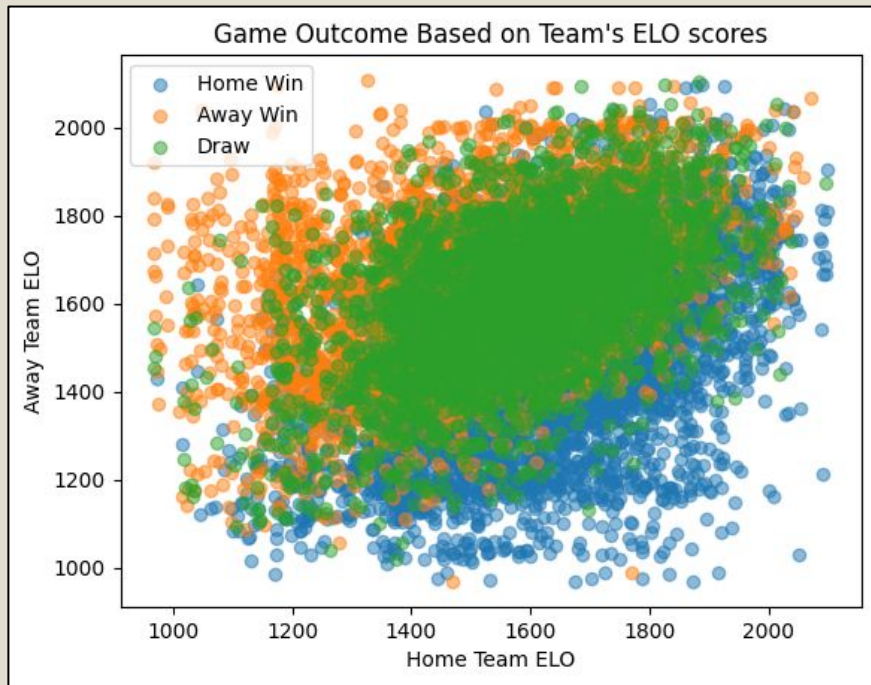


Class 3: Team 2 wins



A perfect **multi classification** problem !

Exploring our Data



Our Models

Baseline

Simple algorithm to get us started.

1

Logistic regression

First version of our advanced ML model.

2

Neural networks

Trying to capture non-linearity to predict better.

3

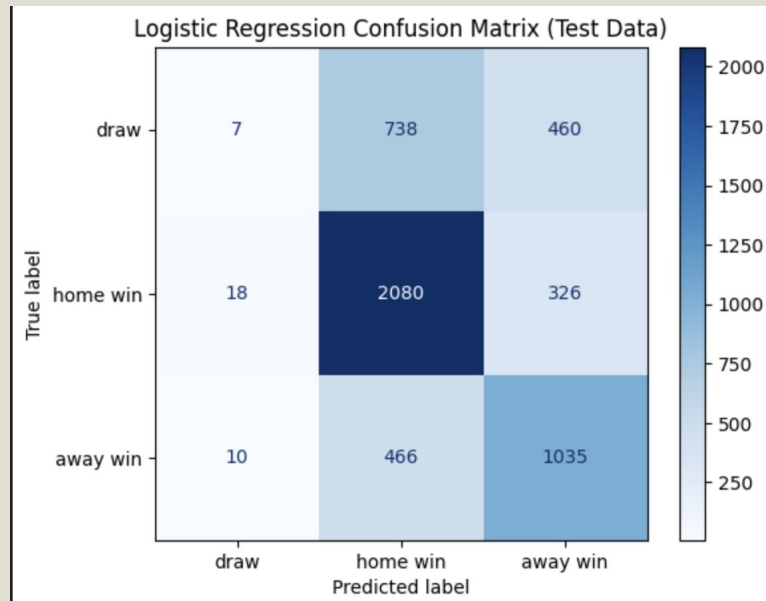
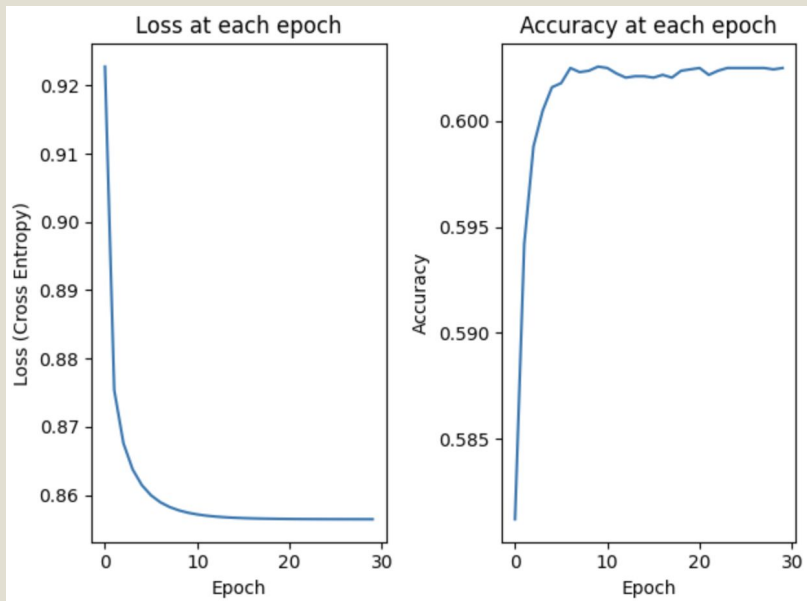
Tree-based algorithm

The industry standard for sports predictions.

4

Baseline + Logistic Regression

- Baseline model test accuracy score: 59.44%
- Logistic Regression model test accuracy score: 60.89%



Neural Networks

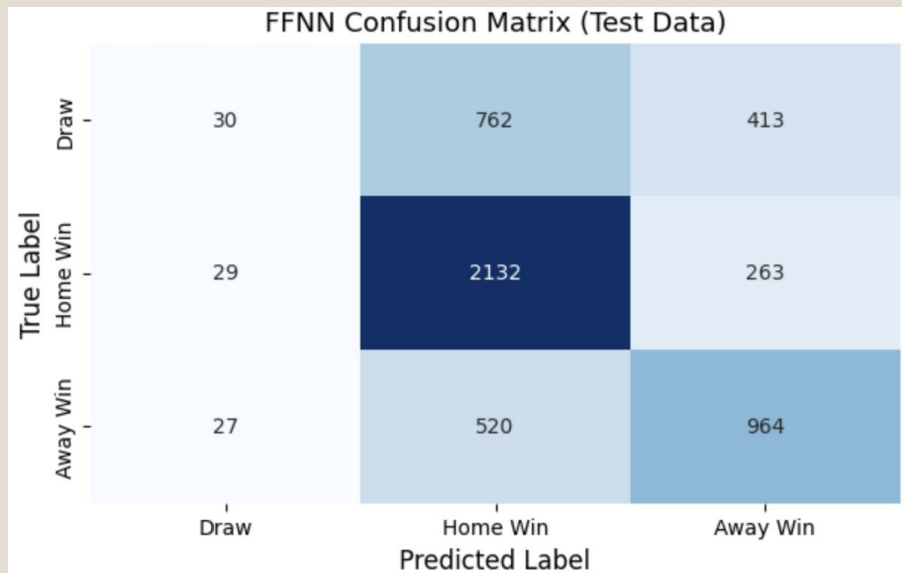
Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 256)	4,096
dense_1 (Dense)	(None, 128)	32,896
dense_2 (Dense)	(None, 64)	8,256
dense_3 (Dense)	(None, 3)	195

Total params: 45,443 (177.51 KB)

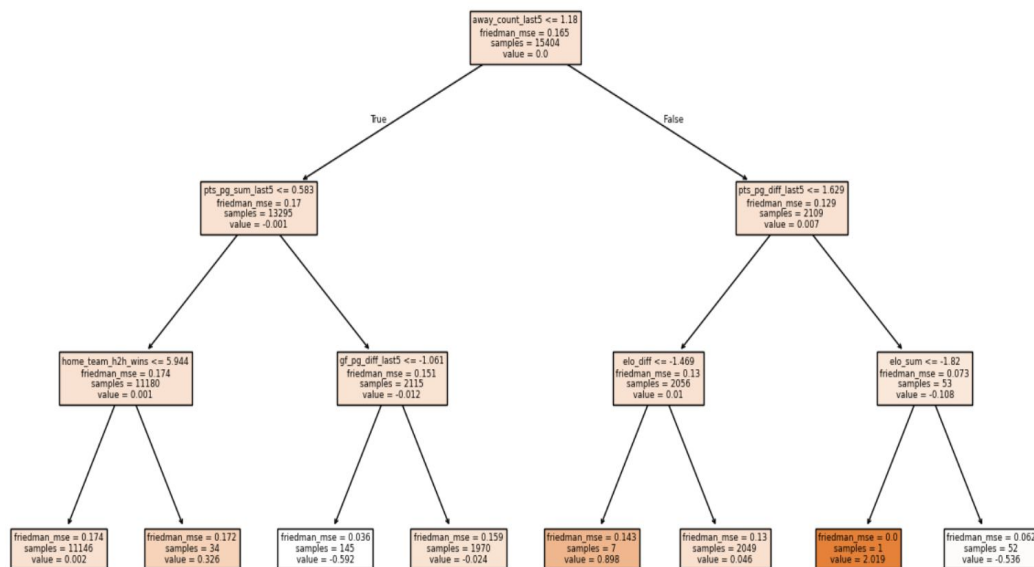
Trainable params: 45,443 (177.51 KB)

Non-trainable params: 0 (0.00 B)

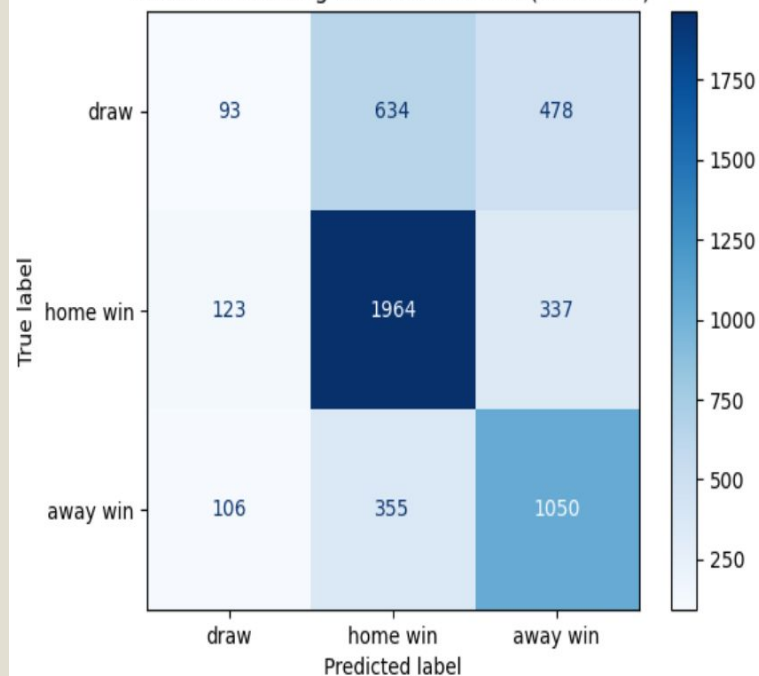


Tree-Based Models

Example Tree from Gradient Boosting



GradientBoosting Confusion Matrix (Test Data)



Model Comparison

Model	Training Accuracy	Validation Accuracy	Test Accuracy	Draw Recall
Baseline	55.15	55.76	59.44	0
Logistic Regression	60.25	60.79	60.89	0.006
Neural Network	60.26	60.62	60.82	0.025
Tree-Based (Gradient Boosting)	62.95	60.83	60.58	0.077

2026 World Cup Group Stage Predictions

GROUP A

1  **MEX** 9 pts

2  **KOR** 6 pts

GROUP B

1  **CAN** 9 pts

2  **SUI** 6 pts

GROUP C

1  **BRA** 9 pts

2  **MAR** 6 pts

GROUP D

1  **USA** 9 pts

2  **AUS** 6 pts

GROUP E

1  **GER** 9 pts

2  **ECU** 6 pts

GROUP F

1  **NED** 9 pts

2  **JPN** 6 pts

GROUP G

1  **BEL** 9 pts

2  **IRN** 6 pts

GROUP H

1  **ESP** 9 pts

2  **URU** 6 pts

GROUP I

1  **FRA** 9 pts

2  **SEN** 6 pts

GROUP J

1  **ARG** 9 pts

2  **ALG** 6 pts

GROUP K

1  **POR** 9 pts

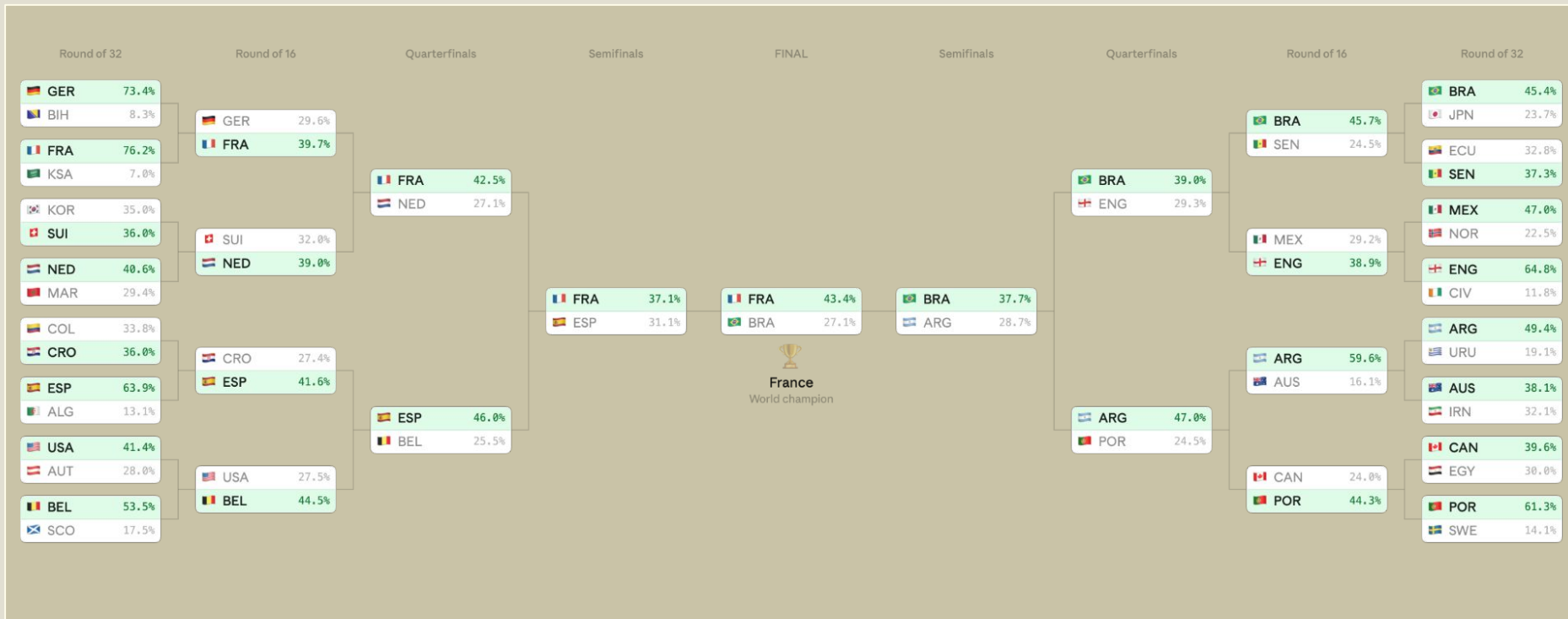
2  **COL** 6 pts

GROUP L

1  **ENG** 9 pts

2  **CRO** 6 pts

2026 World Cup Knockout Stage Predictions



Conclusion

- Improved slightly over baseline accuracy
- All models performed equally in terms of accuracy
 - Possible ceiling on accuracy due to Draw class
 - Many factors that determine winner
- Deploy Gradient Boosting model



Thank you!